

ROBOT FOR RISKY INDOOR NAVIGATION

Students from an engineering institute have developed the prototype of an unmanned ground vehicle that can navigate across territories in an emergency response operation. Different sensors can be plugged in it as per the user's requirements



Indoor navigation robot

■ JALAJA RAMANUNNI

The 2008 terror attack on Mumbai's Taj was fresh in everybody's mind when budding engineers at M.V.J. College of Engineering were brainstorming over what to make for their college project. They decided to design a robot that could assist rescue teams in navigating more efficiently and with lesser risks in an emergency response situation. Thus a theory was developed, tested and transferred to an autonomous surveillance robot.

The aim of their project was to build a robot that could perform



various tasks in place of a soldier during indoor combat operation. It would incorporate video and audio surveillance, and transmit sensitive information to the base station. It would also be used to draw a map to provide the trajectory of its path and remotely acquire the data using wireless communication technology.

The work was divided among the group members according to each

person's area of expertise. Three of them worked on mechanical fixtures, two on programming, two on circuit design and one oversaw the entire project.

Hardware and software used

The robot has two layers—the base layer consists of a battery, motors and sensors, while the top layer consists of a multi-functional board, sensors, wireless router, camera and microphone. The device has open channels for 48 sensors and the prototype uses six at a given time.

"Since National Instruments (NI) allowed us to use their hardware and software, we used the LabVIEW platform, which provided an amazing front-end user panel that could display the calibrated results in a graphical user-interface termed as G-language," shares team member S. Arun Kumar. The language enabled faster programming of the robot. With other languages it would have taken much longer for creating pages of codes required for the device. The code written in LabVIEW could easily be deployed to the sBRIO-9642 hardware using WiFi connectivity.

Kumar adds, "The sBRIO multi-functional board allows manipulation of analogue and digital frequencies. We also used NI's software like NI Robotics module, NI FPGA and NI RT. A customised motor driver circuit was developed by the team to drive